

Weihua Xiao

Tel: (+86) 19821285670, (+1) 9297179833
Email: wx2356@nyu.edu **GitHub:** github.com/FCHXWH823
Location: Austin, Texas, United States

Research Interests: LLM4EDA, logic synthesis, approximate computing

Education

Postdoctoral Researcher, The University of Texas at Austin Nov. 2026 – Present
Research Directions: Large Language Model (LLM)-aided Electronic Design Automation (EDA)
Advisor: David Z. Pan (Fellow of ACM, IEEE, and SPIE; Professor, Silicon Labs Endowed Chair)

Postdoctoral Researcher, New York University Aug. 2024 – Oct. 2026
Research Directions: Large Language Model (LLM)-aided Electronic Design Automation (EDA)
Advisor: Ramesh Karri (Chair & Professor, IEEE Fellow), Siddharth Garg (Professor)

PhD, University of Michigan–Shanghai Jiao Tong University Joint Institute Sep. 2019 – June 2024
Major: Electronic Science and Technology **Advisor:** Weikang Qian (Associate Professor)

Visiting Scholar, University of Alberta, CA Jan. 2023 – Dec. 2023
Advisor: Jie Han (Professor)

Bachelor, Xidian University, CN Oct. 2015 – June 2019
Major: Communication Engineering **Overall GPA:** 3.71/4.0

Publications

I published **nine** papers in **Top Conferences** as the **first/co-first author**:

1. **Weihua Xiao**, Zeng Wang, Minghao Shao, Raghu Vamshi Hemadri, Chang Meng, Johann Knechtel, Muhammad Shafique, Ozgur Sinanoglu, Siddharth Garg, and Ramesh Karri, "TrojanLoC: LLM-based Framework for RTL Trojan Localization and Classification," *Accepted as a regular paper, Proc. IEEE/ACM ICCAD 2026.* **[CCF-B]**
2. **Weihua Xiao**, Jason Blocklove, Johann Knechtel, Ozgur Sinanoglu, Kanad Basu, Jeyavijayan Rajendran, Siddharth Garg, Ramesh Karri, "Special Day – GUIDE: GenAI Units In Digital Design Education," *Proc. DATE 2026*, Verona, Italy. **[CCF-B]**
3. Buddhi Perera*, Zeng Wang*, **Weihua Xiao***, Mohammed Nabeel, Ozgur Sinanoglu, Johann Knechtel, Ramesh Karri, "Focus Session: LLM4PQC – Accurate and Efficient Synthesis of PQC Cores by Feedback-Driven LLMs," *Proc. DATE 2026*, Verona, Italy. *(*Equal contribution.)* **[CCF-B]**
4. **Weihua Xiao**, Derek Ekberg, Siddharth Garg, Ramesh Karri, "Hybrid-NL2SVA: Integrating RAG and Finetuning for LLM-based NL2SVA," *Proc. ACM/IEEE MLCAD 2025*, Santa Cruz, CA.
5. **Weihua Xiao***, Tingting Zhang*, Xingyue Qian, Jie Han, Weikang Qian, "Efficient Approximate Decomposition Solver using Ising Model," *Proc. ACM/IEEE DAC 2024*, San Francisco, CA. **Acceptance Rate: 23%.** *(*Equal contribution.)* **[CCF-A]**
6. **Weihua Xiao**, Shanshan Han, Yue Yang, Shaoze Yang, Cheng Zheng, Jingsong Chen, Tingyuan Liang, Lei Li, Weikang Qian, "MiniTNtk: An Exact Synthesis-based Method for Minimizing Transistor Network," *Proc. IEEE/ACM ICCAD 2023*, San Francisco, CA. **Acceptance Rate: 23%.** **[CCF-B]**
7. **Weihua Xiao**, Weikang Qian, "ASPPLN: Accelerated Symbolic Probability Propagation in Logic Network," *Proc. IEEE/ACM ICCAD 2022*, San Francisco, CA. **Acceptance Rate: 22.5%.** **[CCF-B]**
8. **Weihua Xiao**, Cheng Zhuo, Weikang Qian, "OPACT: Optimization of Approximate Compressor Tree for Approximate Multiplier," *Proc. DATE 2022*, virtual. **Acceptance Rate: 25%.** **[CCF-B]**
9. **Weihua Xiao**, Weikang Qian, Weiqiang Liu, "GOMIL: Global Optimization of Multiplier by Integer Linear Programming," *Proc. DATE 2021*, virtual. **Acceptance Rate: 23.9%.** **[CCF-B]**

Other papers (not first author):

1. Saideep Sreekumar, Zeng Wang, Akashdeep Saha, **Weihua Xiao**, Minghao Shao, Muhammad Shafique, Ozgur Sinanoglu, Ramesh Karri, and Johann Knechtel, "TrojanGYM: A Detector-in-the-Loop LLM for Adaptive RTL Hardware Trojan Insertion," *Accepted as a regular paper, Proc. ACM/IEEE MLCAD 2026.* **Regular-paper Acceptance Rate: 21.1% (37/175 valid submissions; 183 total submissions).**

2. Patrick Yubeaton, Andre Nakkab, **Weihua Xiao**, Luca Collini, Ramesh Karri, Chinmay Hegde, Siddharth Garg, "VeriThoughts: Enabling Automated Verilog Code Generation using Reasoning and Formal Verification," *NeurIPS 2025.* **Acceptance Rate: 24.91%.** **[CCF-A]**
3. Yi Wu, Chuangtao Chen, **Weihua Xiao** et al., "A Survey on Approximate Multiplier Designs for Energy Efficiency: From Algorithms to Circuits," *ACM TODAES*, 2023. **[CCF-B]**
4. Chenfei Lou, **Weihua Xiao**, Weikang Qian, "Quantified Satisfiability-based Simultaneous Selection of Multiple Local Approximate Changes," *Proc. IEEE ISCAS 2022*, Austin, TX. **[CCF-B]**
5. Chen Wang, **Weihua Xiao**, John P. Hayes, Weikang Qian, "Exploring Target Function Approximation for Stochastic Circuit Minimization," *Proc. IEEE/ACM ICCAD 2020*, virtual. **Acceptance Rate: 27.0%.** **[CCF-B]**

Papers under review:

1. Zeng Wang*, **Weihua Xiao***, Minghao Shao*, Raghu Vamshi Hemadri, Ozgur Sinanoglu, Muhammad Shafique, Ramesh Karri, "VeriDispatcher: Multi-Model Dispatching through Pre-Inference Difficulty Prediction for RTL Generation Optimization." *(*Equal contribution.)*
2. Ruogu Chen, **Weihua Xiao**$^{\#}$, Ramesh Karri, Jie Han, "CoEvoP&R: Co-Evolving Placement Objectives with Routing Feedback via Large Language Models." *arXiv:2607.17398.* ($^{\#}$Corresponding author)
3. Chang Meng, **Weihua Xiao**$^{\#}$, Wayne Burleson, Giovanni De Micheli, "Gradient Estimation of Approximate Multipliers for High-Accuracy Deep Learning Model Retraining." ($^{\#}$Corresponding author)

Summary of Research

1. LLM-aided EDA for RTL Synthesis, Verification, and Security (Postdoc Period)

- **Application:** Automates design-space exploration, RTL generation, assertion generation, and security analysis for complex hardware systems.
- **Proposed Work:**
  - **RTL synthesis/generation:** Contributes to *VeriThoughts*, the first synthetic RTL dataset-generation pipeline and first reasoning LLMs for RTL generation; **NeurIPS'25** [**CCF-A**].
  - **RTL synthesis/generation:** Proposes *PrefixLLM*, an LLM-agentic flow for prefix circuit synthesis with SPCR representation and iterative tool feedback; achieves up to 26.71% area reduction; **DAC'25 WIP** [**CCF-A venue**].
  - **RTL verification:** Proposes *Hybrid-NL2SVA* with RAG + fine-tuning for LLM-based SystemVerilog assertion generation; improves functionality match by 58.42% over GPT-4o-mini; **MLCAD'25** (GitHub).
  - **RTL generation optimization and security:** Proposes *VeriDispatcher*, a multi-model dispatching method for RTL generation based on pre-inference difficulty prediction; **DAC'26 WIP** [**CCF-A venue**]. Also proposes LLM-based RTL Trojan localization and classification (*TrojanLoC*); **ICCAD'26** [**CCF-B**].

2. Approximate Computing for Energy-Efficient AI Hardware (PhD Period)

- **Motivation:** AI systems are dominated by matrix and tensor operations; carefully controlled arithmetic approximation can reduce energy while preserving model accuracy through retraining or workload-aware error budgeting.
- **Proposed Work:**
  - Proposes ILP-based synthesis of approximate multipliers co-optimizing error and power/area; achieves avg. 24.4% PDP and 8.4% mean error distance (MED) reduction over prior art; **DATE'22** [**CCF-B**] (GitHub).
  - Proposes Ising model-based approximate disjoint decomposition; achieves 11% MED reduction and 1.16× speedup over state-of-the-art for generic circuit approximation; **DAC'24** [**CCF-A**].
  - Co-authored comprehensive survey "Approximate Multiplier Designs for Energy Efficiency" (*ACM TODAES*, 2023) [**CCF-B**].
  - Ongoing work under review: gradient estimation of approximate multipliers for high-accuracy deep learning model retraining, connecting circuit-level approximation with end-to-end ML accuracy recovery (corresponding author).

3. Logic and Transistor-Level Synthesis for Efficient VLSI Design (PhD Period)

- **Application:** Optimized standard cells, arithmetic blocks, and logic networks are the physical building blocks of energy-efficient AI accelerators and emerging computing systems.
- **Proposed Work:**
  - First formulates transistor network synthesis as SAT; reduces transistor count by up to 9.39% over all 4-input P-class Boolean functions; **ICCAD'23** [**CCF-B**] (GitHub).
  - Global ILP-based multiplier optimization; reduces PDP by up to 71% over industry designs; **DATE'21**

[**CCF-B**] (GitHub).
– Linear-complexity symbolic probability propagation algorithm; improves switching activity estimation accuracy by 24.70% with 29× speedup; **ICCAD'22** [**CCF-B**].

Projects

1. ASIC Standard Cell Customization Oct. 2022 – June 2023

Overview: Extended standard cell library for improved hardware performance and lower power, demonstrating a complete ASIC design flow from synthesis to layout.

Flow:

- Extract new standard cells from digital circuits via Frequent Subgraph Mining (FSM).
- Synthesize minimal transistor networks for new cells using **MiniTNtk** (ICCAD'23), achieving minimum transistor count over benchmarks.
- **Automatically generate hardware layout** for extended standard cells and **extract parasitic parameters**.
- Perform logic synthesis over EPFL benchmarks with the extended library; reduce area by up to **12.6%**.

Selected Honors & Achievements

Google Cloud Research Credits Program (**$5,000**) 2026
Full Membership in *Sigma Xi, The Scientific Research Honor Society* 2026
DATE 2026 Young People Programme 2026
CCF Integrated Circuit Design Thesis Incentive Program (**Nomination**) 2025
National Mathematics Competition of College Students, CN (**First Prize**) 2017

Other Experiences

1. Teaching Experiences:

- 2024.9–2025.6: **Build Your ASIC**, NYU Undergraduate Course (role: **co-instructor**)
- 2025.9–2025.12: **Generative AI Based Chip Design I**, NYU Undergraduate & Master Course (role: **co-instructor**, course website)
- 2026.2–2026.5: **Generative AI Based Chip Design II**, NYU Undergraduate & Master Course (role: **co-instructor**)

2. Academic Activity Organization:

- 2026.5: **AI-based Hardware Attacks (AHA!) Challenge** at 2026 IEEE International Symposium on Hardware Oriented Security and Trust (HOST) (link)
- 2026.7: **GREAT Workshop** (Gen-AI Red-Team Evaluation of Microelectronics Assurance and Trust) at ACM/IEEE Design Automation Conference (DAC 2026), Long Beach, CA (link)
- 2026: **Session Co-Chair**, IEEE/ACM International Conference on Computer-Aided Design (ICCAD)

3. Review Experiences: DAC, ICCAD, DATE, TCAD, TVLSI, JCST

4. Technical Program Committee Member:

- Design, Automation and Test in Europe Conference (DATE): 2027
- IEEE/ACM International Conference on Computer-Aided Design (ICCAD): 2026, 2025
- IEEE International Conference on Computer Design (ICCD): 2025

Technical Skills

- **EDA & ASIC Design:** Standard cell library development, transistor-network synthesis, cell-level layout generation, parasitic extraction, technology mapping, logic synthesis (ABC, Yosys), formal verification
- **Hardware Design:** ASIC design flow (RTL → synthesis → layout), Boolean satisfiability (SAT)-based exact synthesis, ILP-based circuit optimization, Verilog/SystemVerilog
- **AI/ML & LLM:** LLM fine-tuning (LoRA/QLoRA), RAG frameworks, formal verification-guided code generation, model deployment
- **Programming:** Python, C++, MATLAB; SAT solvers (MiniSat), ILP solvers (Gurobi)